

The Importance of Being Nuclear: Designing for Seismic Safety on the Lebanese Coast

Probabilistic Seismic Hazard and Base-Isolation Assessment for Nuclear Power Plants

Abstract

This study develops a preliminary seismic-safety framework for prospective nuclear infrastructure along the Lebanese coast by coupling regional probabilistic seismic hazard analysis (PSHA) with fixed-base and base-isolated structural assessment of a published reference nuclear power plant. R-CRISIS was used to calculate hazard at 714 grid points using the Chiou-Youngs 2008 (CY2008) and Akkar-Bommer 2010 (AB2010) ground-motion models, with a 10 km source-depth case adopted as the principal model and 5 km and 15 km cases retained as sensitivities. Hazard at a representative coastal test location, Site S1 near Naqoura, was then applied to separate two-dimensional North-South (NS) and East-West (EW) OpenSeesPy models of the Auxiliary Shield Building, Steel Containment Vessel, and Containment Internal Structure. The reproduced fixed-base modal periods agree with the reference model within approximately 2.3%. The governing fixed-base response is the EW AB2010-D10 case, with a CQC base shear of approximately 14,389 kip. A 273-bearing lead-rubber bearing (LRB) isolation system was calibrated around a 2.5 s effective period and 20% equivalent damping. The isolated periods are 2.489 s and 2.491 s in the NS and EW directions, respectively, with approximately 99.99% first-mode horizontal effective mass. Effective-linear isolation reduces base shear by approximately 65.5-73.3%, internal structural deformation by 90.5-95.3%, and response-spectrum pseudo-acceleration by 94.1-98.1%, while concentrating about 1.85-1.95 cm of displacement in the isolation layer. Three paired ESM records were also selected and scaled as an exploratory nonlinear-analysis extension, but nonlinear plant-response results are not reported here. The study demonstrates a coherent hazard-to-isolation workflow, while remaining explicitly a screening-level assessment rather than a nuclear licensing-grade site or bearing design.

Keywords: probabilistic seismic hazard analysis (PSHA); Lebanon; nuclear power plant; seismic isolation; lead-rubber bearing (LRB); R-CRISIS; OpenSeesPy; response-spectrum analysis.

1. Introduction

In a previous study in The Importance of Being Nuclear series, titled The Importance of Being Nuclear: A Quantitative Assessment of Lebanon's Energy Crisis, the potential contribution of nuclear power to Lebanon's electricity system was examined from an energy-supply perspective. That study indicated that nuclear generation could, in principle, provide a substantial portion of the country's electricity demand. Energy sufficiency alone, however, does not establish the technical feasibility of nuclear deployment. Nuclear power plants are safety-critical facilities for which external hazards and site conditions must be evaluated with substantially greater rigor than for conventional infrastructure, consistent with international nuclear site-evaluation guidance [8,9]. In Lebanon, one of the most important of these external hazards is earthquake ground motion.

Lebanon lies within the tectonically active eastern Mediterranean and is strongly influenced by the Dead Sea Transform fault system and associated regional seismic sources. The country and surrounding region have experienced damaging historical earthquakes, while both onshore and offshore seismic sources contribute to the potential for future strong ground shaking. For nuclear facilities, this hazard is particularly important because earthquake-induced demand can affect not only the structural system itself but also safety-related equipment, containment functions, cooling systems, and the ability of the plant to maintain a safe condition. Any serious consideration of nuclear power in Lebanon must therefore be accompanied by a quantitative characterization of seismic hazard and an evaluation of structural strategies capable of reducing earthquake demand.

The siting problem is also closely related to the cooling requirements of nuclear generation. Large nuclear power plants require access to a reliable heat sink capable of removing substantial quantities of thermal energy during normal operation and supporting safety-related heat removal. Lebanon has limited inland surface-water resources, many of which are already committed to irrigation, potable-water supply, hydropower production, and other uses. Under a multi-unit nuclear-development scenario, dependence on these relatively limited inland water resources would therefore impose major siting and resource-allocation constraints. The Mediterranean Sea provides a substantially larger and more reliable heat sink and

consequently represents the most practical basis for preliminary nuclear siting in Lebanon. This consideration motivates the concentration of the present seismic investigation on the Lebanese coastal region.

A central component of this study is therefore the development of a probabilistic seismic hazard analysis (PSHA) framework specifically intended for coastal nuclear-siting screening. The hazard model was implemented in R-CRISIS over a computational grid comprising 714 calculation points, allowing the spatial variation of seismic demand to be evaluated across the study region rather than at a single preselected location. Seismic hazard at each point was calculated by integrating earthquake occurrence rates, seismic-source geometries, magnitude distributions, source-to-site distances, and ground-motion prediction relationships.

To address epistemic uncertainty in ground-motion prediction, two independent ground-motion prediction equations were employed: Chiou and Youngs [3] (CY2008) and Akkar and Bommer [4] (AB2010). The two models were run separately rather than blended into a single prediction, permitting their hazard estimates to be compared directly. For regional screening, the resulting hazard values were compared on a point-by-point basis so that the more conservative ground-motion estimate could be identified, while both GMPE branches were retained for the more detailed structural assessment of the selected site. This approach allows uncertainty between the two ground-motion models to remain visible rather than being concealed within a single averaged hazard estimate.

Uncertainty in source depth was examined separately. A 10 km source-depth model was adopted as the principal case, with otherwise identical calculations repeated at 5 km and 15 km to assess the sensitivity of predicted hazard to this assumption. An additional source-model uncertainty was considered for the Ghab fault, for which alternative high- and low-seismicity parameterizations were available for the same source geometry. These alternatives were evaluated in separate calculations and combined without simultaneously activating both representations, thereby avoiding double counting of the same fault source. Hazard outputs were evaluated at the engineering reference level corresponding to a 10% probability of exceedance in 50 years, approximately equivalent to a 475-year return period.

The resulting hazard field was then used to identify a representative coastal location, Site S1 near Naqoura, for detailed structural investigation. The objective of this selection is not to propose or certify a final nuclear site, but to provide a consistent location at which the connection between regional PSHA and nuclear structural response can be examined quantitatively. Site-specific response spectra extracted from the hazard model are subsequently used as seismic input for both fixed-base and base-isolated structural evaluations.

The structural component of the study is based on a published reference nuclear power plant model comprising the Auxiliary Shield Building (ASB), Steel Containment Vessel (SCV), and Containment Internal Structure (CIS). Separate horizontal structural models are reproduced and validated against published modal properties before being assembled into a common nuclear-island representation. The validated fixed-base structure is then subjected to site-specific response-spectrum analysis to establish baseline seismic demands, including base shear, structural deformation, modal response, and acceleration-related demand.

Seismic base isolation is subsequently investigated as a strategy for reducing the transmission of earthquake motion into the nuclear structures. Lead-rubber bearing (LRB) systems with target isolation periods of 2.0, 2.5, and 3.0 s are initially examined through equivalent-linear screening. These alternatives represent the fundamental trade-off associated with isolation design: increasing the isolation period generally reduces transmitted acceleration and force but increases displacement demand. The screening calculations therefore provide a rational basis for selecting an isolation configuration rather than adopting a bearing system independently of the local seismic environment. In the present study, the 2.5 s system is retained as the principal configuration because it provides an intermediate balance between stiffness, transmitted force, and isolation displacement. The preliminary screening explicitly remains an equivalent-linear approximation rather than a substitute for nonlinear bearing response.

A site-specific bilinear LRB system is then calibrated using the long-period seismic demand obtained from the S1 hazard spectra. The principal configuration consists of 273 bearings supporting the combined superstructure and basemat mass, with a target effective isolation period of 2.5 s and an equivalent isolation damping ratio of approximately 20%. The resulting isolated structural system is evaluated through eigenvalue analysis and effective-linear response-spectrum analysis, allowing its modal characteristics and seismic demands to be compared directly with those of the validated fixed-base structure.

Accordingly, the purpose of this study is not to produce a licensing-grade design for a Lebanese nuclear power plant or to establish the final suitability of any particular coastal site. Rather, it investigates whether a technically coherent pathway can be developed linking coastal nuclear siting, probabilistic seismic hazard, nuclear structural response, and seismic isolation design. The study combines regional PSHA, alternative GMPE formulations, source-model and depth sensitivity, fixed-base structural validation, site-specific LRB calibration, isolated modal analysis, fixed-versus-isolated response-spectrum comparison, and exploratory real-record selection and scaling for a later nonlinear extension. Full nonlinear plant response-history results, three-dimensional torsional and rocking response, soil-structure interaction, detailed bearing instability mechanisms, and formal nuclear licensing qualification remain outside the reported scope.

By integrating seismic-hazard characterization with structural and isolation engineering, this study provides a quantitative first-step framework for evaluating the seismic feasibility of future nuclear infrastructure along the Lebanese coast. More broadly, it extends the question of whether nuclear power could contribute to Lebanon's energy system beyond electricity production alone, demonstrating that the viability of nuclear energy must also be examined through the site-specific environmental and engineering constraints required to achieve an acceptable level of seismic safety.

2. Literature Review

The present study is built primarily on two complementary bodies of previous work. The first concerns the characterization of seismic hazard in Lebanon, particularly the development of probabilistic seismic-source models following improved understanding of the Lebanese and offshore tectonic systems. The second concerns the application of elastomeric seismic isolation to nuclear power plants, including the structural representation of nuclear facilities, lead-rubber bearing behavior, and the numerical assessment of isolated nuclear structures. Together, these works provide the geological and structural foundations required to connect site-specific seismic hazard in Lebanon with a preliminary seismic-isolation assessment for nuclear infrastructure.

2.1 Seismic Hazard Assessment in Lebanon

The seismic-hazard component of this research is based primarily on the Lebanese hazard framework developed by Huijer, Harajli, and Sadek [1], which evaluated the implications of the offshore thrust fault system for national seismic hazard. A later re-evaluation by the same authors [2] further updated the Lebanese seismic-hazard model and presented PGA and spectral-acceleration information for major coastal cities. These studies describe Lebanon within the Dead Sea Transform tectonic system and identify important seismic sources including the Mount Lebanon Thrust, Yammouneh, Rachaya, Serghaya, Dead Sea, and Ghab faults, together with surrounding regional sources.

The study emphasized the importance of combining historical and instrumental earthquake records with source geometry, earthquake recurrence, magnitude limits, source-to-site distance, site conditions, and ground-motion attenuation when estimating seismic hazard. These parameters form the fundamental basis of Probabilistic Seismic Hazard Analysis (PSHA).

An important aspect of the Lebanese hazard work was the gradual refinement of the seismic-source model. Earlier models represented Lebanon using relatively broad regional sources, whereas later models explicitly incorporated individual faults and improved knowledge of offshore seismic structures. This does not make the earlier simplified models unusable; rather, the different models represent different levels of detail appropriate to different engineering purposes.

Huijer et al. [1] also considered modern ground-motion prediction relationships, including Akkar and Bommer [4] and Chiou and Youngs [3], and presented probabilistic hazard results in terms of both peak ground acceleration and spectral acceleration. Their work therefore provides the main seismic-source, recurrence, and ground-motion framework on which the PSHA component of the present study is based.

2.2 Seismic Isolation of Nuclear Power Plants

The structural and isolation component of this research is based primarily on the MCEER technical report by Kumar, Whittaker, and Constantinou [5], Seismic Isolation of Nuclear Power Plants using Elastomeric Bearings. The report investigates the use of low-damping rubber and lead-rubber bearings for reducing earthquake demand on nuclear structures and develops numerical models suitable for analyzing isolated nuclear power plants.

Seismic isolation works by introducing a flexible layer between the foundation and the superstructure, thereby increasing the dominant structural period and reducing the horizontal inertial forces transmitted to the nuclear structures. The report documents previous applications of nuclear seismic isolation, including operating plants in France and South Africa, demonstrating that isolation can be implemented at the scale of a complete nuclear facility.

Particular attention was given to lead-rubber bearings (LRBs) because they combine the flexibility of laminated rubber with energy dissipation produced by yielding of the lead core. Kumar et al. report that horizontal damping associated with LRB hysteresis can typically fall within approximately 10–30% of critical damping, allowing the isolation system to reduce seismic forces while maintaining a restoring capability. [5]

The report also provides the reference nuclear structural model adopted in the present study. The nuclear island is represented using three primary structural components: the Auxiliary Shield Building (ASB), Steel Containment Vessel (SCV), and Containment Internal Structure (CIS). These were represented using lumped-mass stick models and reproduced in OpenSees, with modal results checked against the corresponding reference SAP2000 models.

For the isolated configuration, Kumar et al. investigated systems with different isolation periods and bearing strengths, including configurations with nominal periods of approximately 2 and 3 seconds. Their work provides the principal reference for the structural properties, bearing layout, modal validation, and isolation concepts used in the present research.

The report also identifies important limitations of simplified isolation models. Under severe earthquake loading, phenomena such as lead-core heating, buckling, cavitation, axial-load interaction, torsion, and rocking may influence the response. More detailed three-dimensional models are therefore required when individual bearing demands or extreme nonlinear behavior are of interest. These findings provide an important basis for defining the limits of the simplified isolation assessment conducted in this study. [5]

2.3 Basis for the Present Study

The two references address different parts of the same engineering problem. The Lebanese seismic-hazard work provides the basis for estimating earthquake demand, while the MCEER study provides the structural and seismic-isolation framework for a nuclear power plant.

The present research connects these two foundations by applying a Lebanese PSHA framework to a coastal nuclear-siting context and using the resulting seismic demand to evaluate the fixed-base and base-isolated response of a reference nuclear power plant. Rather than directly adopting the seismic demand or bearing properties from the MCEER study, the isolation system is evaluated using the hazard calculated for the selected Lebanese coastal site.

This integration of Lebanese seismic hazard assessment and nuclear seismic-isolation engineering forms the principal basis of the present study.

3. Probabilistic Seismic Hazard Analysis Methodology

3.1 Study Framework and Computational Grid

A Probabilistic Seismic Hazard Analysis (PSHA) was performed using R-CRISIS 20.3.0.0 [6] to estimate earthquake ground-motion hazard across the Lebanese coastal study region. The analysis combines earthquake occurrence rates, seismic-source geometry, magnitude-frequency distributions, source-to-site distance, and ground-motion prediction equations (GMPEs) to calculate the probability of exceeding specified levels of ground motion.

A rectangular computational grid was defined between approximately 33.05°N and 34.70°N latitude and 35.00°E and 36.00°E longitude. A spacing of 0.05° was used in both directions, giving 21 longitude points and 34 latitude points, for a total of 714 calculation sites. The grid was intended for regional coastal screening rather than final nuclear-site qualification.

The approximately 10 km inland coastal corridor used later for regional statistics is therefore a nominal screening corridor rather than a geometrically exact strip along the entire national coastline. In the extreme northern segment, where the coastline approaches the eastern edge of the computational domain (36.00°E), the grid can truncate the full inland width. Coastal statistics are consequently based on the intersection of the nominal 10 km corridor with the available R-CRISIS grid. This assumption is retained explicitly so that the northern coastal statistics are not interpreted as complete coverage of every point within exactly 10 km of the shoreline.

3.2 Seismic-Source Model

The seismic-source model was based on the established tectonic framework used in previous Lebanese seismic-hazard studies. The sources included:

- Mount Lebanon Thrust
- Yammouneh Fault
- Rachaya Fault
- Serghaya Fault
- Zrariye–Chabriha Fault
- Dead Sea Fault
- Ghab Fault (high/low)
- Cyprus region
- Syria region
- remaining/background Lebanon region

Major faults were represented as line sources, while the Cyprus, Syria, and remaining Lebanon regions were represented as area sources. Most of the source traces and area boundaries were manually approximated in Google Earth from the adopted tectonic model and entered into R-CRISIS using longitude-latitude-depth coordinates. Because of this manual digitization, the geometries are treated as approximate representations suitable for regional screening rather than precisely surveyed fault traces. The complete coordinates used in R-CRISIS should therefore be reported in an appendix rather than in the main methodology.

A uniform source depth of 10 km was used for the base model. Two additional cases, 5 km and 15 km, were analyzed with the other model parameters unchanged to assess the sensitivity of the calculated hazard to the assumed source depth.

3.3 Magnitude-Frequency Model

Earthquake recurrence was represented using a modified Gutenberg-Richter relationship. For each source,

$$\lambda(M_0) = 10^{(a - bM_0)}, \quad \beta = b \ln(10)$$

where M_0 is the lower magnitude threshold, a and b are the Gutenberg-Richter recurrence parameters, $\lambda(M_0)$ is the annual occurrence rate of earthquakes with magnitude greater than or equal to M_0 , and β is the corresponding natural-logarithmic slope parameter.

Magnitude calculations were discretized at intervals of 0.1 magnitude units:

$$N_{\text{mag}} = [(M_{\text{max}} - M_0) / 0.1] + 1$$

where M_{max} is the adopted maximum magnitude for the source.

An engineering lower-magnitude threshold of approximately $M_w=5.0$ was used for the main analysis. Events below this level were excluded from the principal hazard model. Consequently, the Syria regional source, whose adopted M_{max} was only 4.6, was retained in the database but remained inactive in the main PSHA.

The remaining Lebanon/background source was intended to represent an effective magnitude of approximately $M_w=5.0$. R-CRISIS did not permit identical minimum and maximum magnitudes, so it was represented using the narrow range $M_0=4.9$, $M_{\text{max}}=5.0$.

This was a numerical workaround required by the software rather than a separate physical magnitude range, and it was therefore documented explicitly.

For the Ghab Fault (high), which has a M_{min} of 6 M_w , the M_0 used for it was $M_w=6$.

The complete seismicity parameters for all sources—source type, M_0 , λ , β , M_{max} , and number of magnitude calculation points—are reported in a separate model-input table.

3.4 Ground-Motion Prediction Equations

Two GMPEs were evaluated independently:

- Chiou and Youngs (2008), CY2008
- Akkar and Bommer (2010), AB2010

The use of two models was intended to account for uncertainty in ground-motion prediction. Rather than combining both equations into a single prediction from the beginning, each GMPE was run separately and the resulting hazard values were compared.

Several methodological decisions used in this procedure were developed with guidance from Prof. Mohamed Harajli, including the independent treatment of the two GMPEs and the use of the larger, more conservative hazard value at each grid point for the final regional screening surface. For this simplified regional analysis, both GMPEs were applied across the modeled source-to-site range up to the adopted 200 km integration limit. This is a screening assumption; model-specific applicability warnings were retained rather than treated as evidence of exact validity over every magnitude-distance combination.

For CY2008, the principal settings were:

- $VS_{30}=760$ m/s
- VS_{30} determination: inferred
- $Z_1(m)=0$
- aftershock: false
- sigma truncation: 0
- maximum integration distance: 200 km
- units coefficient: 1
- acceleration output in g

For AB2010, rock conditions (Rock Vs30>750m/s) were adopted, corresponding to the rock-site formulation of the model. Because the R-CRISIS implementation uses acceleration units of cm/s², a units coefficient of 981 was specified for conversion to g, using

$$1g \approx 981 \text{ cm/s}^2.$$

For simplification, a strike-slip mechanism was used where a mechanism input was required. This represents a modeling simplification and does not imply that every source in the regional tectonic system physically has a strike-slip mechanism.

3.5 Treatment of Ghab-Source Uncertainty

Two alternative seismicity parameter sets were available for the Ghab Fault: Ghab High and Ghab Low. Because they represent alternative characterizations of the same physical fault geometry, they were not activated simultaneously.

For each GMPE and source-depth case, one PSHA run was performed with Ghab High and another with Ghab Low. The corresponding hazard values were then averaged point-by-point.

This procedure, also adopted following methodological guidance from Prof. Harajli, preserves the uncertainty associated with the two Ghab recurrence models while avoiding double-counting of the same fault.

3.6 R-CRISIS Analysis Settings and Calculation Procedure

The principal R-CRISIS settings were:

Parameter	Adopted value
Software	R-CRISIS 20.3.0.0
Exposure period	50 years
Spectral period	T=0 s
Hazard measure	PGA
Output units	g
Minimum acceleration	0.01 g
Maximum acceleration	1.0 g
Number of intensity levels	20
Intensity spacing	Logarithmic
Base source depth	10 km
Sensitivity depths	5 and 15 km

At T=0 s, the calculated spectral acceleration corresponds to peak ground acceleration (PGA).

The averaging of the Ghab Fault (high and low) produced one Ghab-averaged hazard surface for each GMPE, CY2008 and one for AB2010 at each assumed depth. The two GMPE surfaces were then compared at every computational point, and the larger PGA was selected as the conservative screening value.

The process was repeated for source depths of 5, 10, and 15 km, with the 10 km case used as the principal model and the other two cases used to evaluate depth sensitivity.

3.7 Hazard Extraction and Regional Interpretation

R-CRISIS generated hazard curves for all 714 grid points. PGA corresponding to a 10% probability of exceedance in 50 years, approximately equivalent to a 475-year return period, was extracted from the hazard curves.

The full grid was used to create regional hazard surfaces. For a simpler interpretation of variation along the coast, representative points were also selected from the southern, central, and northern coastal regions. Following the regional comparison, a representative coastal site was selected for the subsequent response-spectrum and nuclear structural analyses.

3.8 Methodological Assumptions, Review, and Limitations

The PSHA was developed as a simplified regional screening analysis, not as a licensing-grade nuclear seismic hazard assessment. Several assumptions were therefore required, including uniform source depths within each case, reference rock conditions, simplified fault mechanisms, approximate Google Earth source geometries, and the absence of detailed local soil amplification.

The selected 10% probability of exceedance in 50 years (approximately 475-year return period) is an engineering screening level retained for consistency with the regional comparison; it is not presented as a nuclear design-basis or licensing hazard level. A nuclear site evaluation would require a substantially more rigorous site-specific seismic-hazard program, including comprehensive geological, geophysical, geotechnical, seismological, and uncertainty evaluations consistent with IAEA guidance [8,9].

The Ghab-source uncertainty was treated using High/Low averaging, while uncertainty associated with ground-motion prediction was examined through separate CY2008 and AB2010 calculations. Sources below approximately $M_w=5$ were excluded from the principal engineering model, which resulted in the Syria source remaining inactive. The $M_w=4.9$ – 5.0 representation of the remaining Lebanon source was introduced specifically to overcome the R-CRISIS restriction on equal minimum and maximum magnitudes.

R-CRISIS also generated applicability warnings for some AB2010 magnitude combinations. In particular, the Ghab High model extended beyond the nominal magnitude range of the GMPE, while the inactive Syria source fell below its lower magnitude range. These warnings were retained and documented as limitations of the simplified implementation.

Several of these methodological assumptions were developed with guidance from Prof. Mohamed Harajli, particularly the separate treatment of the two GMPEs, use of the conservative point-by-point result, treatment of the Ghab High and Low alternatives, approximate distance applicability, and exclusion of magnitudes below $M_w=5$. The $M_w=4.9$ – 5.0 background-source workaround was introduced because of the R-CRISIS software limitation and was retained after its justification was documented.

Following completion of the analysis, the calculated hazard values and their regional trends were reviewed with Prof. Harajli. The results were considered reasonable for the adopted level of simplification, broadly consistent with earlier Lebanese seismic-hazard studies, and within the expected uncertainty associated with PSHA. This review is treated as methodological expert feedback rather than formal independent validation of the hazard model.

4. Structural and Seismic-Isolation Methodology

4.1 Reference Nuclear Structural Model

The structural model used in this study was based on the nuclear power plant model presented by Kumar, Whittaker, and Constantinou [5]. The reference plant consists of three main structural components:

- Auxiliary Shield Building (ASB),
- Steel Containment Vessel (SCV),
- Containment Internal Structure (CIS).

The geometric, mass, material, and stiffness properties of these components were taken from the reference report and reproduced in OpenSeesPy using the OpenSees framework [7].

The original reference model contains three-dimensional information; however, the available data were not sufficient to define the superstructure torsional stiffness confidently. For this reason, the present study used separate two-dimensional models in the North–South and East–West directions rather than introducing unsupported assumptions into a full three-dimensional model.

The NS and EW models were first constructed in the fixed-base condition. Their natural periods were compared with those reported in the reference study before any isolation system was introduced. This step was used to confirm that the structural model reproduced the main dynamic behavior of the published nuclear model.

4.2 Fixed-Base Modal Analysis

After the fixed-base model was validated, a modal analysis was performed to determine the natural periods, mode shapes, and horizontal effective modal mass.

A sufficient number of modes was included so that the cumulative horizontal effective mass exceeded 90% in each direction. Modes with less than approximately 1% individual horizontal effective-mass participation were still included when required to reach this cumulative criterion. Exact R-CRISIS spectral periods were obtained for the more significant modes, while spectral values for the lower-participation modes were obtained by interpolation.

The site-specific AB2010 and CY2008 response spectra obtained from the PSHA were then interpolated at the structural modal periods. Modal displacements, pseudo-accelerations, and base-shear contributions were calculated for each mode before the individual modal responses were combined.

Because several structural modes can have similar frequencies, the Complete Quadratic Combination (CQC) method was used as the main modal-combination method. The Square Root of the Sum of Squares (SRSS) method was also calculated as a comparison and numerical check. Code 07 explicitly uses CQC as the final method and SRSS only as an audit comparison.

4.3 Treatment of Damping

The damping treatment required special attention because the PSHA spectra and the nuclear structural model did not originally use the same damping ratio.

The AB2010 and CY2008 spectra generated in R-CRISIS were based on 5% damping, whereas the reference nuclear superstructure model used 4% damping. These two values were not added together. Instead, the positive-period R-CRISIS spectral ordinates were adjusted from 5% to 4% before being applied to the fixed-base structural model.

The damping modification used was

$$\eta = \sqrt{[10 / (5 + \xi)]}$$

where ξ is the required damping ratio expressed as a percentage. Therefore, the 5%-damped spectrum was converted to the 4%-damped structural spectrum before the CQC modal combination was performed. The PGA value at $T=0$ was left unchanged because zero-period ground acceleration is not dependent on oscillator damping.

A python code checks that both the spectrum used for the structural response and the CQC modal combination correspond to the same 4% damping level.

A separate damping assumption was later introduced for the isolation system. A 20% equivalent damping ratio was used for the dominant LRB isolation response. This represents the energy dissipation provided by the lead-rubber bearings and is separate from the 4% damping of the superstructure.

Thus, the three damping values have different roles.

4.4 Preliminary Isolation-System Screening

The isolation system was based on lead-rubber bearings (LRBs), following the general configuration and bearing concepts presented in the reference nuclear-isolation report.

Three target isolation periods were considered: 2.0 s, 2.5 s, 3.0 s.

These were used as preliminary screening cases rather than three complete isolation designs.

For each target period, the required effective horizontal stiffness of the isolation system was estimated from the total mass supported by the bearings:

$$\omega = 2\pi/T \quad K_{eff, total} = M \omega^2.$$

The required total stiffness was then distributed among the identical bearings adopted from the reference plant layout.

The original 5%-damped site spectrum was also converted to an equivalent 20%-damped isolation spectrum for this screening stage using the same damping-modification relationship. This was only an equivalent-linear approximation intended to represent the reduced response caused by LRB hysteretic energy dissipation.

The three target periods were compared in terms of the general trade-off between stiffness, seismic force, and displacement. The 2.5-s system was retained as the principal isolation period because it represented an intermediate solution between the stiffer shorter-period system and the more flexible longer-period system.

4.5 Preliminary LRB Model

After selecting the principal target period, an individual lead-rubber bearing was represented using a simplified bilinear force-displacement model.

The model consisted of:

- an initial stiffness K_1 ,
- a post-yield rubber stiffness K_d ,
- a characteristic lead strength Q_d ,
- an effective stiffness K_{eff} at the design displacement.

The effective stiffness was represented as

$$K_{eff} = K_d + Q_d/D.$$

The post-yield-to-initial stiffness ratio was defined as $\alpha = K_d/K_1$. The reference report did not provide one unique numerical α value that could be transferred directly to the simplified two-dimensional bearing model. Therefore, $\alpha = 0.10$ was adopted as the principal modeling assumption, while $\alpha = 0.05$ and 0.15 were retained as sensitivity cases.

The bearing was calibrated so that the bilinear model provided approximately 20% equivalent damping while maintaining the selected effective isolation period.

The hysteretic energy dissipated by the idealized bearing was represented by

$$ED = 4Q_d(D - D_y), \text{ and the corresponding equivalent damping was calculated from } \xi_{eff} = ED / (2\pi K_{eff} D^2).$$

The calibration procedure adjusted the characteristic strength of the bearing until the required equivalent damping was reached while maintaining positive post-yield stiffness and ensuring that yielding occurred before the design displacement.

Physical bearing properties such as rubber-layer configuration, lead yield stress, cover thickness, steel shims, and the number of bearings were taken from the reference nuclear-isolation model rather than independently designed in this study.

The resulting bearing should therefore be interpreted as a preliminary calibrated LRB model, not as a final manufacturing or qualification design.

4.6 Isolated Structural Model

The validated fixed-base model was then modified by introducing the isolation layer beneath the common basemat.

Because the analysis was performed separately in two dimensions and the basemat was treated as rigid in horizontal translation, the individual identical bearings were represented by an equivalent aggregate horizontal spring. In this simplified directional model, the aggregate spring represents the combined stiffness of the parallel bearings beneath the nuclear island.

Two stiffness conditions were checked:

- the effective stiffness representing the isolation system at its design response level;
- the larger initial stiffness representing the bearing before significant yielding.

The modal analysis of the isolated model was then used to confirm that the principal horizontal response had shifted from the short-period structural modes toward the longer isolation mode.

This check was important because the main purpose of base isolation is for the nuclear island to move largely together on the flexible isolation layer while reducing deformation within the structures above it.

4.7 Effective-Linear Isolated Response-Spectrum Analysis

The final isolation assessment in this study used an effective-linear response-spectrum analysis.

The dominant isolation mode was evaluated using 20% equivalent damping, while the higher structural modes retained the 4% superstructure damping.

Because the isolation mode and the higher structural modes have different damping ratios, they were treated as separate response groups:

- higher structural modes were combined using CQC;
- the isolation-mode response and the combined structural-mode response were then combined using SRSS.

This separation is explicitly implemented in the isolated response-spectrum procedure; the 20% isolation damping and 4% structural damping are not summed.

The isolated analysis considered quantities such as:

- isolation-layer displacement,
- structural deformation relative to the moving basemat,
- structural pseudo-acceleration,
- and total base shear.

An important distinction was made between total displacement relative to the ground and structural deformation relative to the basemat. A base-isolated structure is expected to experience noticeable motion of the basemat while reducing deformation of the structures located above it.

The isolated results were then compared with the fixed-base results to evaluate the effectiveness of the proposed LRB system.

4.8 Real-Record Selection and Scaling for Nonlinear Extension

Real-record ground motions were prepared as an exploratory extension for subsequent nonlinear response-history analysis. Three paired horizontal records were selected from the Engineering Strong Motion Database (ESM) [10]. Because the PSHA was not deaggregated into a controlling magnitude-distance-epsilon scenario, the record search used broad screening ranges rather than claiming a unique controlling earthquake: strike-slip mechanism, $M_w = 5.5-7.5$, $V_{s30} = 600-900$ m/s, and epicentral distance = 5-100 km. Each record retained its two processed horizontal accelerograms, with HNN assigned to the NS model and HNE assigned to the EW model.

The original ESM accelerograms were used to compute 5%-damped pseudo-acceleration spectra. For each earthquake pair, one scalar factor was applied to both horizontal components so that their original relative amplitude relationship was preserved. Separate factors were calculated for AB2010-D10 and CY2008-D10 by matching the geometric mean of the two horizontal spectra exactly to the corresponding Site S1 target at $T = 2.5$ s, the principal isolation period. The entire 0.4-3.0 s spectral range was retained as an audit. This is intentionally an isolation-period-focused scalar scaling procedure, not a broadband spectral-matching or nuclear qualification procedure.

The scaled records are preparation inputs only. The effective-linear response-spectrum results remain the final plant-response results reported in this paper; nonlinear response histories must be separately verified before they can be incorporated into the conclusions.

4.9 Main Assumptions and Limitations

The structural and isolation analysis was intended as a preliminary engineering assessment rather than a complete nuclear qualification study.

The main assumptions were:

- structural properties were taken from the Kumar et al. reference model;
- separate 2D NS and EW models were used instead of a full 3D model;
- the basemat was treated as rigid for the directional isolation analysis;
- identical LRBs were assumed beneath the basemat;
- individual bearings were represented by an equivalent aggregate horizontal isolation element;
- 4% damping was used for the nuclear superstructure;
- 20% equivalent damping was assumed for the main isolation response;
- a bilinear LRB model was used;
- the post-yield stiffness ratio was treated as an explicit modeling assumption and checked through sensitivity cases;
- soil-structure interaction was not included;
- torsional and full three-dimensional bearing effects were not evaluated;
- detailed buckling, cavitation, heating, and vertical bearing behavior were outside the scope of the effective-linear model;
- nonlinear response-history results were not used as part of the final reported plant response; only real-record selection and scaling were completed as preparation.

Therefore, the isolation results represent the expected effective-linear horizontal behavior of the proposed system and should not be interpreted as final bearing qualification or licensing-level nuclear seismic design.

5. Results and Discussion

5.1 PSHA Results

5.1.1 Seismic-Source Geometry

Seismic source	Type	Coordinates used	No. of vertices
Ghab Fault	line	(36.264,34.703),(36.282, 35.82)	2
Dead Sea Fault	line	(35.4, 31),(35.6, 33.3)	2
Mount Lebanon Thrust	line	(35.215, 33.517), (35.06, 33.814), (35.378, 33.965), (35.573, 34.274), (35.741, 34.416)	5
Yammouneh Fault	line	(35.56, 33.27),(35.704, 33.617),(35.807, 33.839),(35.992, 34.052),(36.178, 34.237),(36.267, 34.387),(36.265, 34.645)	7
Rachaya Fault	Line	(35.68, 33.291),(35.764, 33.437),(	5

Seismic source	Type	Coordinates used	No. of vertices
		35.813, 33.513),(35.906, 33.547),(35.96, 33.605)	
Serghaya Fault	line	(35.808, 33.504),(36.004, 33.529),(36.101, 33.685),(36.116, 33.782),(36.156, 33.839),(36.254, 33.991),(36.315, 34.077),(36.398, 34.143),(36.499, 34.186),(36.604, 34.189)	10
Zrariye–Chabriha Fault	line	(35.211, 33.289),(35.31, 33.306),(35.359, 33.373),(35.404, 33.43)	4
Lebanon Remaining	Area	(34.702, 33.123),(35.462, 32.966),(35.563, 33.307),(36.111, 33.246),(37.079, 34.902),(35.219, 34.664)	6
Cyprus	Area	(32.861, 35.582),(32, 35.94),(30.355, 35.813),(31.518, 34.122),(34.856, 34.324),(35.594, 36.25)	6
Syria region	Area	(36.468, 33.766),(36.366, 33.429),(37.112, 33.221),(37.349, 33.653),(36.56, 33.93)	5

Table 1. Seismic-source geometries used in the R-CRISIS model.

Note: For area sources, boundary vertices are listed in anticlockwise order. Coordinates are given as (longitude, latitude). For line sources, vertices are listed sequentially along the fault trace.

The seismic-source geometries used in the R-CRISIS model are summarized in Table 1. The fault traces and area-source boundaries were represented using the longitude–latitude coordinates described in the methodology.

5.1.2 Regional PGA Distribution

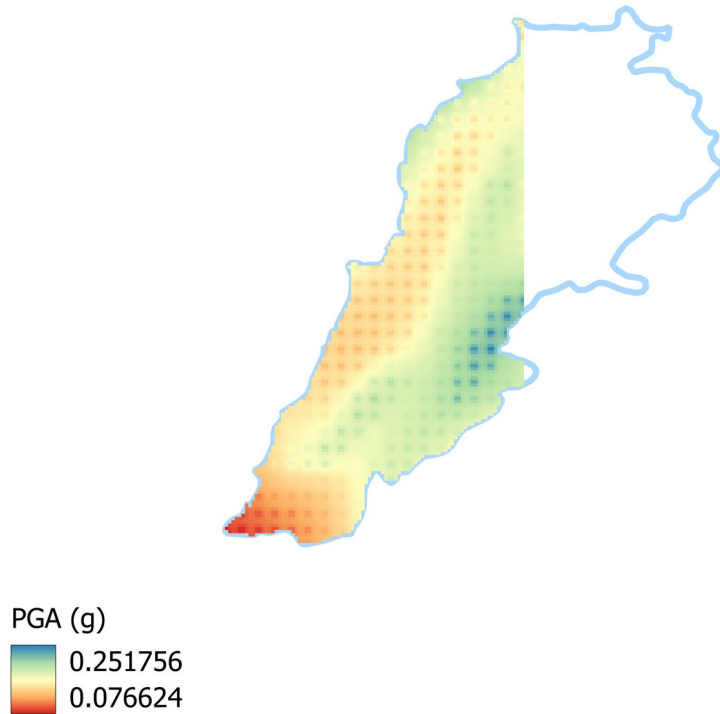


Figure (1) (a): D10 conservative PGA

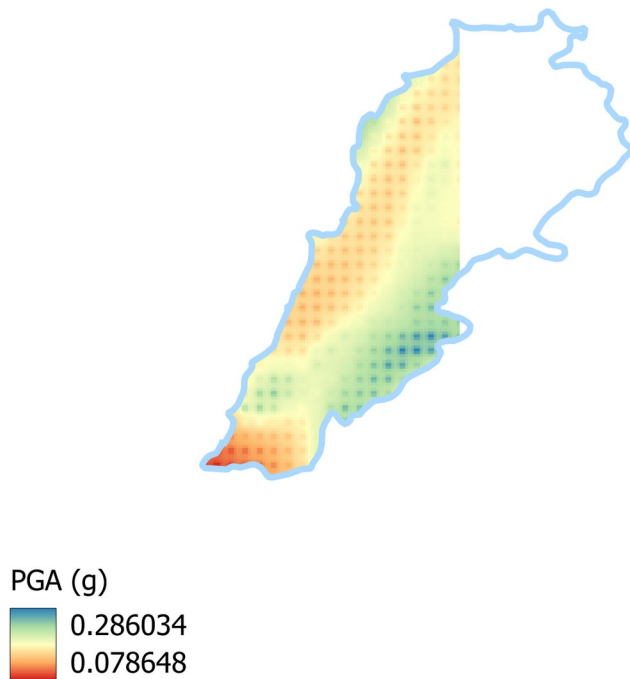


Figure (1) (b): D5 conservative PGA

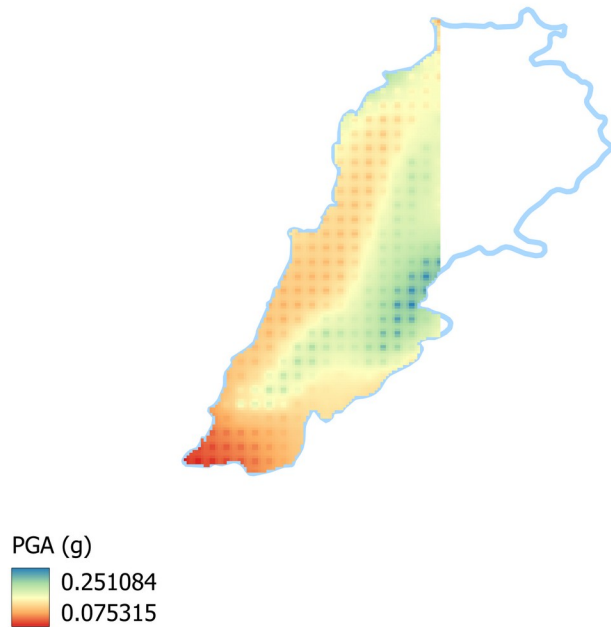


Figure (1) (c): D15 conservative PGA

Figure 1. Spatial distribution of conservative PGA corresponding to a 10% probability of exceedance in 50 years for assumed source depths of (a) 10 km, (b) 5 km, and (c) 15 km. At each computational grid point, the larger PGA obtained from AB2010 and CY2008 was retained.

The three conservative PGA maps show the spatial variation of seismic hazard across the computational domain for the D5, D10, and D15 source-depth assumptions. The geographical distribution of hazard remains generally similar among the three cases, although differences in PGA magnitude occur with changing source depth.

5.1.3 Coastal PGA Statistics

Depth	Minimum PGA (g)	Mean PGA (g)	Maximum PGA (g)
5 km	0.07865	0.16937	0.24518
10 km	0.07662	0.15256	0.20832
15 km	0.07531	0.14323	0.20642

Table 2. Coastal conservative PGA statistics for the three source-depth cases.

Verification: Lebanon 0–10 km inland coastal corridor



Figure 2. Nominal coastal screening corridor extending approximately 10 km inland from the Lebanese coastline, intersected with the available R-CRISIS computational grid.

To summarize seismic hazard along the Lebanese coast, computational grid points falling within the nominal approximately 10 km inland corridor were used. Because the R-CRISIS domain terminates at 36.00°E, this corridor is clipped where the extreme northern coastline approaches the eastern grid boundary. The minimum, mean, and maximum PGA values in Table 2 therefore describe the available coastal-grid sample rather than an exact, uniformly complete 10 km strip along every part of the coastline. Figure 2 illustrates this screening domain and Table 2 reports the corresponding D5, D10, and D15 statistics.

The coastal statistics in Table 2 show a systematic but moderate sensitivity to the assumed source depth. The mean conservative PGA within the approximately 10 km coastal corridor decreases from 0.169 g for D5 to 0.153 g for D10 and 0.143 g for D15. Relative to the D10 base case, the D5 mean is about 11.0% higher, whereas the D15 mean is about 6.1% lower. The maximum coastal PGA is more sensitive on the shallow side, increasing from 0.208 g at D10 to 0.245 g at D5, an increase of about 17.7%, while the D15 maximum remains close to the D10 value at 0.206 g. These results indicate that source-depth uncertainty changes the level of predicted coastal hazard more than it changes the overall spatial pattern. The 10 km model was therefore retained as the principal case, with the 5 km and 15 km models interpreted as sensitivity bounds rather than independent preferred hazard models.

5.1.4 Site S1 Hazard and GMPE Comparison

Site S1, located near Naqoura, was selected as a representative coastal location for coupling the regional PSHA with the nuclear structural model. The site is used as an analytical test location rather than as a proposed or qualified nuclear site. The site-specific PGA values extracted from the two GMPE branches are summarized in Table 3.

Table 3. Site S1 PGA values for the two GMPE branches and three source-depth assumptions.

Source depth	AB2010 PGA (g)	CY2008 PGA (g)	Conservative PGA (g)
5 km	0.05350	0.07745	0.07745
10 km	0.05258	0.07619	0.07619
15 km	0.05258	0.07473	0.07473

At the D10 base depth, CY2008 gives 0.07619 g compared with 0.05258 g from AB2010, so the CY2008 PGA is approximately 44.9% larger. In contrast, changing the source depth at S1 modifies the PGA by only about $\pm 2\%$ around the D10 value for the CY2008 branch and by less than 2% for AB2010. This comparison is important because it shows that, at the selected site, uncertainty associated with the GMPE is materially larger than the sensitivity to the simplified uniform source-depth assumption. Retaining the two GMPE branches separately in the structural analysis was therefore justified: an averaged spectrum would have concealed a meaningful difference in predicted ground motion.

The subsequent structural calculations also show why the controlling GMPE cannot be identified from PGA alone. CY2008 controls the zero-period PGA at S1, yet AB2010 later produces the larger fixed-base structural base shear. After isolation shifts the dominant response to approximately 2.5 s, CY2008 again produces the larger isolated base shear. This change in the controlling branch reflects the period dependence of the response spectra and demonstrates the value of carrying both hazard branches through the structural assessment.

5.2 Validation of the Fixed-Base Nuclear Structural Model

Before using the site-specific spectra, the separate NS and EW OpenSeesPy models were checked against the published modal periods of the ASB, SCV, and CIS reference models. The comparison is given in Table 4. Agreement is close in all six cases, with the largest absolute period difference remaining below about 2.3%.

Table 4. Fixed-base modal validation against the reference nuclear model.

Structure	Direction	Reference T (s)	OpenSeesPy T (s)	Difference
ASB	NS	0.323	0.32982	+2.11%
ASB	EW	0.362	0.36104	-0.27%
SCV	NS	0.180	0.18025	+0.14%
SCV	EW	0.275	0.27535	+0.13%
CIS	NS	0.079	0.07719	-2.29%
CIS	EW	0.086	0.08599	-0.01%

Four of the six comparisons differ by less than 0.3%, while the two largest deviations are approximately 2.11% for the ASB NS model and 2.29% for the CIS NS model. For a reproduced lumped-mass stick model, this level of agreement provides a strong check that the directional mass and stiffness representation is consistent with the reference model. It also supports the decision to use separate two-dimensional NS and EW models rather than introducing an unsupported three-dimensional torsional stiffness.

5.3 Fixed-Base Modal Participation and Site-Specific Seismic Response

Sixty modes were evaluated in each horizontal direction. In the NS model, the cumulative horizontal effective modal mass reached 93.28%, with the 90% threshold first exceeded at mode 43. In the EW model, the cumulative horizontal effective mass reached 93.33%, with the 90% threshold first exceeded at mode 41. The relatively large number of modes required is consistent with the multi-component stick representation of the nuclear island: the ASB, SCV, and CIS contribute distinct short-period modes rather than behaving as a single oscillator in the fixed-base condition.

Table 5. Modal participation retained for the fixed-base response-spectrum analysis.

Direction	Mode at which cumulative mass first exceeds 90%	Final cumulative horizontal mass
NS	43	93.277%
EW	41	93.330%

The 5%-damped R-CRISIS spectra were converted to the 4% damping level used by the reference nuclear superstructure before modal combination. The final fixed-base base shears obtained using CQC are shown in Table 6. AB2010 produces the larger base shear in both directions even though CY2008 produces the larger S1 PGA.

Table 6. Fixed-base CQC base shear at Site S1.

Direction	Hazard branch	Base shear (kip)
NS	AB2010-D10	13,747.9
NS	CY2008-D10	11,724.3
EW	AB2010-D10	14,389.1
EW	CY2008-D10	12,099.7

The governing fixed-base result is the EW AB2010-D10 case, with a base shear of approximately 14,389 kip. The EW base shear exceeds the NS value by about 4.7% for AB2010 and 3.2% for CY2008, demonstrating that the directional structural properties still influence the fixed-base response. More importantly, the fixed-base branch ordering illustrates that PGA should not be used as a substitute for period-dependent structural demand. The nuclear structures respond to spectral ordinates at their modal periods, so a GMPE that is lower at $T = 0$ can still govern the combined structural response.

5.4 Preliminary Isolation-Period Screening

Three target isolation periods, 2.0, 2.5, and 3.0 s, were examined using a 20% equivalent damping ratio for the isolation response. The total isolated mass was 146,284,555 kg and the bearing layout contained 273 LRBs, consistent with the reference nuclear-isolation configuration. The screening results are summarized in Table 7.

Table 7. Effective-linear isolation-period screening results.

Target period (s)	Hazard branch	20%-damped S_a (g)	Spectral displacement (cm)	Equivalent force (kip)
2.0	AB2010-D10	0.01440	1.431	4,645
2.0	CY2008-D10	0.01400	1.391	4,514
2.5	AB2010-D10	0.01202	1.867	3,878
2.5	CY2008-D10	0.01268	1.968	4,088
3.0	AB2010-D10	0.01072	2.396	3,456
3.0	CY2008-D10	0.00843	1.884	2,717

The results show the expected reduction in transmitted force as the isolation period is lengthened, but the displacement response is governed by the detailed spectral shape rather than by period alone. In the AB2010 branch, increasing the target period from 2.5 to 3.0 s reduces equivalent force from approximately 3,878 to 3,456 kip but increases spectral displacement from 1.87 to 2.40 cm. In the CY2008 branch, however, the sharp drop in spectral acceleration near 3.0 s causes the 3.0-s displacement to remain slightly below the 2.5-s value. This branch-dependent behavior shows that isolation-period selection should be based on the site-specific spectrum rather than on a generic assumption that a longer isolation period will always increase displacement.

The 2.5-s configuration was retained as the principal system because it provides an intermediate compromise between the higher force associated with the 2.0-s system and the increased flexibility of the 3.0-s system. For the 2.5-s case, CY2008-D10 controls the preliminary isolation demand, with a 20%-damped spectral acceleration of approximately 0.01268 g and a displacement of 1.968 cm. The 2.0- and 3.0-s cases remain useful as sensitivity checks but are not treated as separate final bearing designs.

5.5 Site-Specific LRB Calibration

Using the controlling 2.5-s CY2008-D10 demand, a bilinear LRB was calibrated to reproduce an effective period of 2.5 s and approximately 20% equivalent damping at the design displacement. The principal model used a post-yield-to-initial stiffness ratio of $\alpha = 0.10$. The resulting per-bearing parameters are summarized in Table 8.

Table 8. Principal site-specific LRB calibration for $\alpha = 0.10$.

Quantity	Calculated value
Effective stiffness, K_{eff}	3.3847 MN/m
Post-yield stiffness, K_d	2.2590 MN/m
Initial stiffness, K_1	22.5902 MN/m
Characteristic strength, Q_d	22.153 kN
OpenSees yield force, F_y	24.615 kN
Yield displacement, D_y	1.090 mm
Calibration displacement	19.680 mm
Q_d / supported weight	0.422%
Calculated lead-core diameter	57.6 mm
Inferred rubber shear modulus	0.400 MPa
Rubber shear strain at calibration	6.35%

The analytical equivalent damping ratio at the calibration displacement was essentially 20%, and the yield displacement was far below the design displacement, confirming that the bilinear element would be operating in its post-yield range at the effective design point. The calculated rubber shear strain of about 6.35% is small for an elastomeric bearing with 310 mm total rubber thickness, which is consistent with the relatively low long-period demand calculated for Site S1.

The most unusual calibration result is the characteristic-strength ratio Q_d/W of approximately 0.42%. Kumar et al. [5] investigated reference nuclear isolation systems with characteristic strengths ranging from about 6% to 18% of bearing gravity weight. The much smaller ratio obtained here should therefore not be interpreted as a generally recommended nuclear LRB strength. It arises from forcing the simplified bilinear model to deliver about 20% equivalent damping at a design displacement of only about 20 mm under the low long-period S1 hazard. The result is best viewed as a site-specific analytical calibration that demonstrates the mechanics of the proposed system, not as a final manufacturable or licensing-qualified isolator.

5.5.1 Sensitivity to the Bilinear Post-Yield Ratio

Table 9. Sensitivity of the calibrated bilinear shape to α .

α	Initial stiffness K_1 (MN/m)	Yield displacement D_y (mm)	Q_d (kN)	Lead-core diameter (mm)
0.05	45.88	0.492	21.46	56.7
0.10	22.59	1.090	22.15	57.6
0.15	14.74	1.842	23.09	58.8

The sensitivity study shows that the same effective period and approximately the same equivalent damping can be reproduced by noticeably different bilinear backbones. Reducing alpha to 0.05 increases the initial stiffness and reduces the yield displacement, while increasing alpha to 0.15 produces a softer pre-yield response and a larger yield displacement. Characteristic strength changes much less. This demonstrates that matching only the effective period and damping at one displacement does not uniquely define the nonlinear force-displacement relation. Accordingly, alpha = 0.10 is retained as the central modeling assumption rather than presented as a uniquely identified physical property.

5.6 Verification of the Isolated Dynamic Model

The calibrated isolation layer was then introduced beneath the common basemat. Because the present analysis is directional and two-dimensional, the 273 identical bearings were represented by one equivalent aggregate horizontal spring. For global translational response of a rigid basemat, the stiffness and strength of identical parallel bearings are additive, so this representation preserves the intended horizontal isolation stiffness without implying that individual-bearing vertical, torsional, or rocking demands are being resolved.

Table 10. Dynamic properties of the isolated model.

Direction	Effective isolated period (s)	Difference from 2.5 s target	1st-mode horizontal mass	Initial-tangent period (s)
NS	2.48926	0.43%	99.9959%	0.97454
EW	2.49061	0.38%	99.9936%	0.97842

The effective isolated periods are 2.489 s in the NS direction and 2.491 s in the EW direction, both within 0.5% of the 2.5-s target. The first isolation mode contains approximately 99.99% of the horizontal effective mass in both directions. This is a fundamental change from the fixed-base condition: rather than distributing the response among many structural modes, the isolated plant responds predominantly as one translating nuclear island on a flexible base.

The initial-tangent periods are much shorter, approximately 0.975-0.978 s. This is physically consistent with the bilinear LRB model because the initial stiffness is substantially larger than the post-yield stiffness. The effective period therefore describes the softened response around the design displacement, whereas the initial-tangent period describes the stiffer system before significant lead yielding. The difference between the two is an expected feature of the chosen bearing model rather than a modeling inconsistency.

5.7 Fixed-Base versus Isolated Response

The final plant-level assessment within the adopted scope used effective-linear response-spectrum analysis. The dominant isolation mode was evaluated with 20% equivalent damping, the higher structural modes retained 4% damping, and the two response groups were combined without summing the damping ratios. Table 11 compares the fixed-base and isolated base shears.

Table 11. Fixed-base and isolated base-shear comparison.

Direction	Hazard branch	Fixed-base shear (kip)	Isolated shear (kip)	Reduction
NS	AB2010-D10	13,747.9	3,846.4	72.02%
NS	CY2008-D10	11,724.3	4,046.7	65.48%
EW	AB2010-D10	14,389.1	3,844.3	73.28%
EW	CY2008-D10	12,099.7	4,045.5	66.57%

Base isolation reduces the calculated base shear by approximately 65.5-73.3% for all four direction-GMPE combinations. The largest reduction occurs for the EW AB2010-D10 case, where the base shear decreases from about 14,389 kip to 3,844 kip. Even the least favorable case, NS CY2008-D10, retains a reduction of about 65.5%. These are substantial global reductions and show that the period shift produced by the isolation layer is effective for both retained hazard branches.

The results also reveal a change in which GMPE governs the structural demand. AB2010 controls the fixed-base base shear, whereas CY2008 produces the larger isolated base shear. This is a direct consequence of moving the dominant response from the short structural periods to the approximately 2.5-s isolation period. The controlling hazard model is therefore response-period dependent. This observation is particularly important for site-specific nuclear assessment because it demonstrates that selecting one GMPE solely from PGA could lead to a non-conservative conclusion for another structural configuration.

Directional differences also become very small after isolation. The NS-EW difference in isolated base shear is less than 0.1% for each GMPE branch, compared with several percent in the fixed-base condition. The near equality of the isolated NS and EW responses is consistent with the modal result showing that essentially the entire plant mass participates in one translational isolation mode. In other words, the isolation layer suppresses much of the direction-specific influence of the superstructure on the global base reaction.

5.7.1 Redistribution of Displacement

Table 12. Effective-linear isolation-layer displacement.

Direction	Hazard branch	Basemat / isolation displacement (cm)
NS	AB2010-D10	1.852
NS	CY2008-D10	1.948
EW	AB2010-D10	1.851
EW	CY2008-D10	1.948

The isolation layer accommodates approximately 1.85-1.95 cm of horizontal displacement. The important result is not the absolute translation alone but where the deformation occurs. The total motion of upper structural control nodes becomes close to the basemat translation, while deformation relative to the moving basemat becomes small. For example, in the NS CY2008-D10 case the basemat translates about 0.767 in. The ASB control node moves about 0.789 in in total but deforms only about 0.0219 in relative to the basemat; the CIS moves about 0.768 in in total while deforming only about 0.00136 in; and the SCV moves about 0.772 in while deforming only about 0.00487 in.

Across the evaluated cases, internal structural deformation is reduced by approximately 90.5-95.3% relative to the fixed-base condition. This illustrates the central physical mechanism of base isolation: earthquake-induced motion is not eliminated but is intentionally concentrated in the isolation interface, thereby reducing distortion of the safety-related structures above it.

5.7.2 Reduction in Structural Pseudo-Acceleration

The same period shift produces very large reductions in response-spectrum pseudo-acceleration. The ASB shows reductions of approximately 95.6-96.6%, the SCV approximately 94.1-95.6%, and the CIS approximately 97.9-98.1%. For the EW AB2010-D10 case, the fixed-base CIS pseudo-acceleration is about 0.654 g and the isolated value is about 0.0121 g, a reduction of approximately 98.15%. In the same case, the ASB decreases from about 0.399 g to 0.0137 g and the SCV from about 0.277 g to 0.0122 g.

For a nuclear facility, acceleration reduction is important because the demand on equipment, supports, piping, and nonstructural components can be strongly acceleration-sensitive. The present values, however, are response-spectrum pseudo-accelerations and must be interpreted accordingly. They are not nonlinear time-history absolute accelerations and should not be presented as such.

5.8 Ground-Motion Suite Preparation for Nonlinear Extension

The ESM search produced three paired records satisfying the adopted broad screening criteria. All six horizontal components passed the metadata audit, and all records use a 0.005 s sampling interval. The selected events span Mw 6.0-6.3, Vs30 values of 767-882 m/s, and epicentral distances of 26.0-81.1 km. Table 13 summarizes the retained suite.

Table 13. Selected paired ESM records for the nonlinear-analysis extension.

Record	ESM event ID	Station	M _w	Vs30 (m/s)	Epicentral distance (km)	NS / EW components
GM1	EMSC-20140203_0000008	KRI1	6.0	882	73.8	HNN / HNE
GM2	EMSC-20150416_0000081	KSS1	6.2	767	26.0	HNN / HNE
GM3	INT-20201229_000001	GOLS	6.3	779	81.1	HNN / HNE

For each hazard branch, the geometric mean of the paired horizontal 5%-damped spectra was anchored exactly to the Site S1 target at 2.5 s. The required scale factors are relatively large, ranging from approximately 3.00 to 3.89 for AB2010-D10 and from 3.16 to 4.10 for CY2008-D10. Table 14 gives the exact values.

Table 14. Scalar scale factors for the three paired records.

Record	AB2010-D10 scale factor	CY2008-D10 scale factor
GM1	3.55936	3.75224
GM2	2.99506	3.15736
GM3	3.89041	4.10123

The spectral audit confirms exact matching at the isolation anchor period, but it also shows that scalar scaling does not create a broadband match. In particular, the three-record suite mean is commonly above the target at shorter periods, especially for CY2008-D10, while some ordinates beyond 2.5 s fall below the AB2010 target. Accordingly, these records are defensible only as an exploratory isolation-focused nonlinear input set. A future production-quality nonlinear study should seek a larger candidate pool and tighter spectral-compatibility criteria to reduce scale factors and improve broadband agreement.

5.9 Overall Discussion

Taken together, the results show that the site-specific LRB concept substantially changes the horizontal dynamics of the modeled nuclear island. The fixed-base structure requires dozens of modes to capture more than 90% of the horizontal effective mass, whereas the isolated configuration concentrates approximately 99.99% of the mass in a single long-period translational mode. This transition from multi-modal structural deformation to global base translation is accompanied by base-shear reductions of approximately 65.5-73.3%, internal deformation reductions of approximately 90.5-95.3%, and pseudo-acceleration reductions of approximately 94.1-98.1%, while the isolation layer accommodates approximately 1.85-1.95 cm of horizontal displacement.

The results also show that the effectiveness of isolation cannot be summarized by one hazard number. CY2008 controls PGA at S1 and the long-period isolated response, whereas AB2010 controls the fixed-base base shear. Similarly, the 3.0-s screening case does not produce a uniform displacement trend across both GMPE branches. These findings reinforce the central methodological choice of retaining period-dependent spectra from both GMPEs instead of selecting a single ground-motion model or relying only on PGA.

At the same time, the LRB calibration identifies an important engineering caution. The very low site-specific long-period demand leads to a calculated Qd/W ratio of only about 0.42%, far below the 6-18% characteristic-strength range examined in the reference nuclear study (Kumar et al., 2015). Therefore, the numerical effectiveness of the isolated structural system is stronger evidence than the physical maturity of the specific bearing calibration. The present bearing should be treated as an analytical prototype whose manufacturability, stability, aging behavior, and qualification would require additional engineering work.

5.10 Limitations of the Reported Results

The strength of the conclusions must remain consistent with the adopted model. The NS and EW directions are evaluated independently using two-dimensional stick models and a rigid basemat. The aggregate isolation element is suitable for global horizontal translation, but it does not resolve coupled bidirectional bearing response, individual-bearing axial-load redistribution, basemat torsion, rocking, or vertical isolation behavior. Kumar et al. [5] similarly note that a full three-dimensional model with explicit bearings is required when individual bearing demands, rocking, and torsional response are of interest.

The hazard model has additional screening-level limitations. The 714-point grid is coarse relative to a site-specific nuclear investigation, source geometries were manually approximated, a uniform source depth was imposed within each case, and the nominal 10 km coastal corridor is clipped at the extreme north by the eastern grid boundary. In addition, the 475-year hazard level is an engineering comparison level rather than a nuclear licensing criterion. The calculated coastal statistics and S1 demand should therefore be interpreted as screening results, not final site characterization [8,9].

The exploratory real-record suite introduces a further limitation. Scale factors of approximately 3.0-4.1 were required to match the 2.5 s target, and the scaled suite is not uniformly compatible with the target over the full 0.4-3.0 s range. Any nonlinear results obtained from this suite should therefore be presented as sensitivity or proof-of-concept results unless the record-selection and spectral-compatibility procedure is strengthened.

The analysis also excludes soil-structure interaction and advanced bearing deterioration mechanisms such as lead-core heating, displacement-dependent buckling, cavitation, and nonlinear vertical behavior. Most importantly, the final reported plant response is effective-linear response-spectrum response. The 20% isolation damping is an equivalent representation of LRB energy dissipation and does not reproduce cycle-by-cycle yielding and unloading.

Final scope boundary: the reported plant-response results intentionally end with the effective-linear isolated response-spectrum analysis. Real-record selection and scalar scaling were completed only to prepare a nonlinear extension. Cycle-by-cycle nonlinear plant response, beyond-design-basis simulations, and conclusions derived from those analyses are not included because verified response-history outputs were not available for this manuscript.

5.11 Principal Findings

- The simplified regional PSHA produces coherent coastal hazard patterns, with source depth affecting the magnitude of PGA more than the overall spatial distribution.
- At Site S1, CY2008 produces substantially larger PGA than AB2010, while the source-depth sensitivity is comparatively small.
- The reproduced fixed-base NS and EW structural models match the published modal periods within approximately 2.3%, supporting their use for the site-specific assessment.
- The governing fixed-base base shear is approximately 14,389 kip for the EW AB2010-D10 case.
- A 2.5-s isolation period provides a balanced principal configuration for the S1 spectrum and was successfully reproduced in the full isolated model.
- The isolated first mode contains approximately 99.99% of the horizontal effective mass, confirming the intended nearly rigid-body translation of the nuclear island on the isolation layer.
- Effective-linear isolation reduces base shear by approximately 65.5-73.3%, internal structural deformation by approximately 90.5-95.3%, and pseudo-acceleration by approximately 94.1-98.1%.
- The controlling GMPE changes after isolation, demonstrating that hazard dominance is period dependent and cannot be inferred from PGA alone.
- The calculated Qd/W ratio of approximately 0.42% is unusually low relative to the reference nuclear-isolation parameter range and requires physical engineering validation before the bearing can be interpreted as a practical design.
- Three paired ESM records were selected and scaled for a nonlinear extension; the required factors (approximately 3.0-4.1) and the imperfect broadband spectral match are material limitations that must be addressed before treating future nonlinear results as production-level evidence.

Overall, the analysis supports the conclusion that long-period base isolation is highly effective for reducing the modeled horizontal seismic demands at Site S1. The strongest evidence is the combined shift of the dominant period to approximately 2.49 s, near-total first-mode mass participation, large reductions in force and internal deformation, and strong reduction in pseudo-acceleration. The study should therefore be interpreted as a technically coherent demonstration of the seismic-isolation concept for a Lebanese coastal nuclear-siting scenario, while stopping short of final bearing qualification or licensing-level nuclear seismic design.

6. Conclusions

This study established a complete screening-level chain from regional seismic hazard to structural response and seismic-isolation assessment for a reference nuclear power plant on the Lebanese coast. The regional PSHA shows a coherent coastal hazard pattern and a moderate sensitivity to the assumed uniform source depth, while the difference between CY2008 and AB2010 is more consequential at Site S1. CY2008 governs PGA at S1, but AB2010 governs the fixed-base base shear, demonstrating that nuclear structural demand cannot be inferred from PGA alone and must be evaluated using period-dependent spectra.

The reproduced nuclear structural model provides a credible basis for this comparison. The six principal ASB, SCV, and CIS directional modal periods agree with the published reference values within approximately 2.3%, and more than 90% cumulative horizontal modal mass is captured in both fixed-base directions. The governing fixed-base result is the EW AB2010-D10 case, with a CQC base shear of approximately 14,389 kip.

The 2.5 s LRB configuration substantially modifies the global dynamics of the nuclear island. The effective isolated periods of 2.489 s and 2.491 s closely reproduce the target, and the first horizontal isolation mode contains approximately 99.99% of the effective mass. Within the adopted effective-linear framework, isolation reduces base shear by approximately 65.5-73.3%, internal structural deformation by 90.5-95.3%, and response-spectrum pseudo-acceleration by 94.1-98.1%, while concentrating approximately 1.85-1.95 cm of displacement at the isolation interface. These results demonstrate the intended mechanism of seismic isolation: the plant moves predominantly as a rigid nuclear island on a flexible base instead of developing large deformation in the safety-related structures above.

The strongest limitations concern the transition from analytical effectiveness to nuclear engineering qualification. The calibrated Q_d/W ratio of approximately 0.42% is far below the characteristic-strength range investigated in the reference nuclear-isolation study [5], and the simplified 2D aggregate-bearing model cannot resolve torsion, rocking, vertical response, individual-bearing axial-load redistribution, heating, buckling, or cavitation. The regional PSHA is also not a licensing-grade nuclear hazard assessment: it uses a 475-year screening level, approximate source geometry, reference rock conditions, and a computational grid whose nominal 10 km coastal corridor is clipped in the extreme north. These constraints do not invalidate the comparative results, but they define the level at which the conclusions can be used.

Overall, the results support the technical plausibility of long-period base isolation as a powerful demand-reduction strategy for a Lebanese coastal nuclear-siting scenario. They do not establish the acceptability of Site S1, a final LRB design, or a licensing basis. The principal contribution of the work is instead the demonstration of a reproducible hazard-to-structure workflow and the identification of the specific uncertainties that must be resolved before the concept can progress from screening research to site-specific nuclear engineering.

7. Recommended Next Steps

- 1. Extend and refine the PSHA domain.** The grid should be expanded eastward in the extreme north and the coastal corridor should be generated using a GIS-based shoreline-distance calculation so that a complete and geometrically exact inland buffer is available along the full coast.
- 2. Upgrade the hazard model toward nuclear practice.** Future work should use a formal logic-tree treatment of epistemic uncertainty, PSHA deaggregation, refined fault and area-source geometries, magnitude-dependent mechanisms, explicit GMPE applicability checks, and site-specific geological/geotechnical data in accordance with nuclear site-evaluation guidance [8,9].
- 3. Evaluate hazard at nuclear-relevant exceedance levels.** The 10% in 50 years result should remain a regional screening benchmark only; site-specific nuclear work should adopt the hazard metrics and probability levels required by the applicable regulatory framework.
- 4. Improve the real-record suite before production nonlinear analysis.** The ESM records selected here [10] are useful for a pilot study, but the approximately 3.0-4.1 scale factors and imperfect broadband match justify a larger candidate search, tighter spectral-compatibility requirements, and, where technically justified, alternative scaling or spectral-matching strategies.
- 5. Complete nonlinear response-history analysis in a staged manner.** First verify a single 1.00x pilot case, including convergence, units, absolute acceleration reconstruction, bearing yielding, and hysteresis. Then run the full three-record, two-direction, two-GMPE suite at 1.00x before considering selected 1.25x and 1.50x beyond-design sensitivity cases.
- 6. Develop a full three-dimensional nuclear-island model when adequate torsional stiffness data become available.** Explicit representation of all bearings is required to resolve basemat torsion, rocking, bidirectional interaction, axial-load redistribution, and individual-bearing demand.
- 7. Perform physical LRB design and qualification checks.** The analytical bearing should be redesigned or rationalized with realistic characteristic strength, displacement capacity, shear strain, lead-core dimensions, buckling stability, cavitation resistance, heating, aging, repeated-cycle behavior, and manufacturability.
- 8. Add soil-structure interaction and equipment-response assessment.** A mature nuclear study should couple the structural model to site-specific soil/rock properties and evaluate floor response spectra, piping, supports, mechanical/electrical equipment, and safety-related systems rather than relying only on global structural response.

8. References

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